CORRECTING AN ELEMENT BY THE RADICAL QUOTIENT TO MAKE A SINGLE GENERATED SUBMODULE A FREE DIRECT SUMMAND

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 39025-39033 (canonical tag 0GVI) and the counted proof is the proof environment at source lines 39035-39105, which immediately follows it and is its own proof.

Editorial note. The proof displays a defect in the author's text at proof lines 39042-39043, where the free direct summand is asserted of M' rather than of the fibre of P; the printed text is copied exactly as it stands and is not corrected, since the lane delivers the author's own words.

Editorial note. The three references are all to labelled results of another chapter: freeness of projectives over local rings, the unit criterion modulo the radical, and finiteness of a Tor group for finite modules over a Noetherian ring. All three are navigation stubs with their roles recorded in the reasons.

Editorial note. The chapter file pinned-original-source.tex carries only an input of the shared preamble on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus the native theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- A finitely generated projective module over a local ring is free, so the localizations of P at maximal ideals are free and the statement makes sense. invoked at proof lines 39036-39037. Makes the rank and freeness language meaningful for the localizations appearing in the hypotheses.
- An element of a ring which is congruent to one modulo the Jacobson radical is a unit. invoked at proof lines 39049-39051. Turns the congruence of the dual functional modulo the radical into a splitting over the whole ring.
- For a finite module over a Noetherian ring the first Tor group against a residue field is finite, and a finite dimensional vector space cannot surject onto a subquotient that has infinite dimension. used at proof lines 39069-39076. Produces an element t independent of the image of s + m in the fibre, which is the input to the next linear algebra step.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original theorem 0593: sources/stacks-chow/tags.txt, source lines 6801–6801. The reference is invoked at proof lines 39036-39037 so that the localizations of P at maximal ideals are free and the statement makes sense; the construction that follows is inside the counted span.

Source reference. Original lemma 0AME: sources/stacks-chow/tags.txt, source lines 13402–13402. The reference supplies the unit criterion modulo the Jacobson radical, used at proof lines 39049-39051 to turn the congruence of the lifted functional into a splitting; the lifting and the choice of m are inside the counted span.

Source reference. Original lemma 0AZ4: sources/stacks-chow/tags.txt, source lines 13812–13812. The reference makes the kernel of the displayed map a quotient of a finite Tor group, used at proof lines 39072-39074 to conclude that the image of K in the fibre has infinite dimension; the linear algebra step built on that claim is inside the counted span.

Lemma 0.2. *Let R be a ring with Jacobson radical J such that R/J is Noetherian. Let P be a projective R-module such that $P_{\mathfrak{m}}$ has infinite rank for all maximal ideals $\mathfrak{m}$ of R. Let $s \in P$ and $M \subset P$ such that $Rs + M = P$. Then we can find $m \in M$ such that $R(s + m)$ is a free direct summand of P.*

Proof. The statement makes sense as $P_{\mathfrak{m}}$ is free by Algebra, Theorem 0593.

Denote $M' \subset P/JP$ the image of M and $s' \in P/JP$ the image of s. Observe that $R/Js' + M' = P/JP$. Suppose we can find $m' \in M'$ such that $R/J(s' + m')$ is a free direct summand of M'. Choose $\varphi' : P/JP \to R/J$ which gives a splitting, i.e., we have $\varphi'(s' + m') = 1$ in R/J. Then since P is a projective R-module we can find a lift $\varphi : P \to R$ of φ'. Choose $m \in M$ mapping to m'. Then $\varphi(s + m) \in R$ is congruent to 1 modulo J and hence a unit in R (Algebra, Lemma 0AME). Whence $R(s + m)$ is a free direct summand of P. This reduces us to the case discussed in the next paragraph.

Assume R is Noetherian. Let $m \in M$ be an element and let $\varphi_1, \ldots, \varphi_n : P \to R$ be R-linear maps. Denote

$$Z(s + m, \varphi_1, \ldots, \varphi_n) \subset \mathrm{Spec}(R)$$

the vanishing locus of $\varphi_1(s + m), \ldots, \varphi_n(s + m) \in R$.

Suppose $\mathfrak{m}$ is a maximal ideal of R and $\mathfrak{m} \in Z(s + m, \varphi_1, \ldots, \varphi_n)$. Set $K = M \cap \bigcap \mathrm{Ker}(\varphi_i)$. We claim the image of

$$K/\mathfrak{m}K \to P/\mathfrak{m}P$$

has infinite dimension. Namely, the quotient P/K is a finite R-module as it is isomorphic to a submodule of $P/M \oplus R^{\oplus n}$. Thus we see that the kernel of the displayed arrow is a quotient of $\mathrm{Tor}_1^R(P/K, \kappa(\mathfrak{m}))$ which is finite by Algebra, Lemma 0AZ4. Combined with the fact that $P/\mathfrak{m}P$ has infinite dimension we obtain our claim. Thus we can find a $t \in K$ which maps to an element $\bar{t}$ of the vector space $P/\mathfrak{m}P$ which is linearly independent of the image $\overline{s + m}$ of $s+m$. By linear algebra, we find an R-linear map $\overline{\varphi} : P \to \kappa(\mathfrak{m})$ such that $\overline{\varphi}(\bar{t}) = 1$ and $\overline{\varphi}(\overline{s + m}) = 0$. Since P is projective, we can find an R-linear map $\varphi : P \to R$ lifting $\overline{\varphi}$. Then we see that the vanishing locus $Z(s + m + t, \varphi_1, \ldots, \varphi_n, \varphi)$ is contained in $Z(s + m, \varphi_1, \ldots, \varphi_n)$ but does not contain $\mathfrak{m}$, i.e., it is strictly smaller than $Z(s + m, \varphi_1, \ldots, \varphi_n)$.

Since $\mathrm{Spec}(R)$ is a Noetherian topological space, we see from the arguments above that we may find $m \in M$ and $\varphi_1, \ldots, \varphi_n : P \to R$ such that the closed subset $Z(s + m, \varphi_1, \ldots, \varphi_n)$ does not contain any closed points of $\mathrm{Spec}(R)$. Hence $Z(s +$

$m, \varphi_1, \ldots, \varphi_n) = \emptyset$. Hence we can find $r_1, \ldots, r_n \in R$ such that $\sum r_i \varphi_i(s+m) = 1$.
Hence

$$R \xrightarrow{s+m} P \xrightarrow{\sum r_i \varphi_i} R$$

is the desired splitting. □